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## RC-139 (Pre-Registration): The 24-Hole Fiber Bundle — Base Space Redefinition

### RESEARCH CYCLE PRE-REGISTRATION: RC-139

**Title:** The 24-Hole Fiber Bundle — Redefining the Base Space After RC-138

**Framework:** 24D-DMF v8.4.3

**Date:** 2026-07-09

**Status:** PRE-REGISTERED — Results Binding Upon Execution

**Dependencies:** RC-138 (fiber bundle failure on 6-axis base, H3/H4 partial successes), RC-137 ( $U_{\theta}$  phase rotation structure), RC-134 (DH23 fixed point, Class D genuine basins), RC-130b (5-fold axis locality), RC-126 (orbit class partition), RC-124 (color sequence injectivity), RC-122 (icosahedron embedding), RC-116 (dimensional cascade), RC-110 (Floquet tick infrastructure)

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## 1. Background

RC-138 tested whether the 6 icosahedral 5-fold axes could serve as the base space of a fiber bundle logical encoding. The verdict was **FAIL**:

- **H1 failed:** The Floquet tick does not permute the 6 axes. Each axis contains deep holes from multiple orbit classes, and the Floquet tick preserves orbit classes, not axes.
- **H2 failed:**  $U_{\theta}$  does not preserve axis assignment for general states constructed as convex combinations of deep holes on the same axis.
- **H3 passed:**  $U_{\theta}$  IS a uniform phase rotation on the fiber coordinate  $\varphi = \arctan2(v[23], v[0])$ .
- **H4 passed:** The Floquet engine corrects “base errors” (noise in coordinates 1–22) with >99.9% probability.

The critical insight from RC-138 is that the **fiber coordinate is well-defined and uniformly rotated by  $U_{\theta}$** , but the **base space was misidentified**. The 6-axis structure is a geometric shadow of the 3D projection, not the dynamical base preserved by the Floquet engine.

RC-139 asks: **What is the correct base space?**

This requires:

1. Identifying a base space that is preserved by the Floquet tick
  2. Verifying that  $U_{\theta}$  acts only on the fiber (not the base) for that base space
  3. Defining logical states as sections of the bundle
  4. Testing whether the resulting encoding supports genuine logical operations
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## 2. Core Questions

| Question                                                   | Why It Matters                                                                                                                                                                                    |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1: Is the 24-deep-hole set itself the base space?</b>  | Each deep hole is a distinct point. The Floquet tick permutes deep holes (within orbit classes). $U_\theta$ preserves nearest deep hole identity. This is the simplest candidate.                 |
| <b>Q2: Are the 5 orbit classes the base space?</b>         | The Floquet tick preserves orbit classes. But $U_\theta$ changes which orbit class a state belongs to (via nearest deep hole). So orbit classes are not preserved by $U_\theta$ .                 |
| <b>Q3: Can a hybrid base space work?</b>                   | Perhaps the base is “orbit class + axis” or some other combined invariant. Or perhaps the fiber bundle is only defined within each orbit class separately.                                        |
| <b>Q4: Does the fiber bundle support a qubit encoding?</b> | A qubit needs two distinguishable logical states. With 24 base points, we have $24^\infty$ possible sections (one phase per point). Can we find two sections that are orthogonal and manipulable? |

### 3. Candidate Base Spaces

#### 3.1 Candidate A: The 24 Deep Holes (Discrete Base)

**Definition:** Base space  $B = \{0, 1, \dots, 23\}$  (one point per deep hole).

**Fiber:**  $F = S^1$  (phase circle, coordinate  $\varphi$ ).

**Bundle:**  $E = B \times F$  with total space of 24 circles.

**Section:**  $|\psi_L\rangle = \{(s, \varphi_s) : s \in \{0, \dots, 23\}\}$ .

**Floquet action:**  $T(t)$  permutes the base points (within orbit classes).

**$U_\theta$  action:**  $U_\theta(s, \varphi_s) = (s, \varphi_s + \theta)$  — acts only on fiber.

**Why it might work:**

- $U_\theta$  preserves nearest deep hole identity (verified in RC-138)
- The Floquet tick permutes deep holes (verified in RC-126)
- Both operations are well-defined on this base

**Why it might not work:**

- 24 base points is large; logical states are high-dimensional
- The Floquet permutation is not a single global permutation but 5 separate permutations (one per orbit class)
- Error correction requires the Floquet dynamics to return to the SAME deep hole, not just the same orbit class

### 3.2 Candidate B: The 5 Orbit Classes (Coarse Base)

**Definition:** Base space  $B = \{A, B, C, D, E\}$ .

**Why it might work:**

- Floquet tick preserves orbit classes (RC-126)
- Natural dynamical partition

**Why it does NOT work:**

- $U_\theta$  changes orbit class membership (verified in RC-138)
- Cannot be a fiber bundle base if  $U_\theta$  acts on it

### 3.3 Candidate C: Hybrid — Orbit Class with Internal Fiber Bundle

**Definition:** Define a separate fiber bundle **within each orbit class**.

- Class A (8 holes): Base = 8 points, Fiber =  $S^1$
- Class B (7 holes): Base = 7 points, Fiber =  $S^1$
- Class C (6 holes): Base = 6 points, Fiber =  $S^1$
- Class D (2 holes): Base = 2 points, Fiber =  $S^1$
- Class E (1 hole): Base = 1 point, Fiber =  $S^1$  (trivial)

**Why it might work:**

- Floquet tick preserves each orbit class (acts as internal permutation)
- $U_\theta$  preserves nearest deep hole within each class (to be verified)
- Each sub-bundle is independently well-defined

**Why it might not work:**

- $U_\theta$  might map between orbit classes (changing nearest deep hole to one in a different class)
- The sub-bundles have different sizes; no uniform logical encoding across all
- Class E (DH23) is trivial; Class D has only 2 points

### 3.4 Candidate D: The 12 Icosahedron Vertices (Intermediate Coarseness)

**Definition:** Group the 24 deep holes into 12 antipodal pairs (vertices of the icosahedron). Base = 12 vertices.

**Why it might work:**

- More natural geometric structure than 24 holes
- Each vertex corresponds to a 5-fold axis direction (but both + and -)

**Why it might not work:**

- The Floquet tick does not preserve antipodal pairs (to be verified)
- $U_\theta$  might not preserve vertex assignment

## 4. Pre-Registered Hypotheses

### H1 — 24-Hole Base is Preserved by Floquet

For any deep hole  $s$ , after one Floquet tick, the state maps to a state whose nearest deep hole is in the same orbit class as  $s$ .

More precisely: The Floquet tick induces a permutation on each orbit class separately.

**Test:** Compute the permutation induced by one Floquet tick on each orbit class.

**Pass if:** Each orbit class is permuted internally (no cross-class transitions).

### H2 — $U_\theta$ Preserves Nearest Deep Hole Identity

For any state  $v$  near deep hole  $s$  (within a neighborhood),  $U_\theta(v)$  has nearest deep hole  $s$ .

**Test:** For each deep hole  $s$  and a grid of phases  $\varphi$ , construct  $v_s(\varphi)$ , apply  $U_\theta$ , verify nearest deep hole is still  $s$ .

**Pass if:** Nearest deep hole is preserved for all  $s$  and all  $\varphi \in [0, 2\pi)$ .

### H3 — Uniform Phase Rotation on 24-Hole Base

For any state  $v_s(\varphi)$  near deep hole  $s$  with fiber angle  $\varphi$ :

$$\varphi(U_\theta(v_s(\varphi))) = \varphi + \theta \pmod{2\pi}$$

**Test:** Measure fiber angle before and after  $U_\theta$  for states near each deep hole.

**Pass if:** Phase shift equals  $\theta$  within  $10^{-6}$  for all  $s$  and  $\varphi$ .

### H4 — Logical Qubit Encoding on 24-Hole Base

Two sections  $|0_L\rangle$  and  $|1_L\rangle$  can be defined such that:

- They are distinguishable (different phase patterns across the 24 holes)
- $U_\theta$  acts as  $R_z(\theta)$  on the logical subspace
- The Floquet tick preserves the logical subspace (maps sections to sections)

**Test:** Define two candidate sections with contrasting phase patterns (e.g., all phases 0 vs. alternating 0/). Verify  $U_\theta$  rotates both uniformly and Floquet tick preserves the section structure.

**Pass if:** Both sections are preserved by  $U_\theta$  and Floquet tick, and they are orthogonal.

### H5 — Orbit-Class Sub-Bundle Structure

Within each orbit class, the Floquet tick induces a cyclic permutation of the deep holes, and  $U_\theta$  preserves the class membership.

**Test:** For each orbit class, compute the Floquet-induced permutation. Verify it is a single cycle (or known cycle structure).

**Pass if:** Each class has a well-defined permutation structure and  $U_\theta$  preserves class membership for states near class holes.

## 5. Falsification Criteria

| Criterion | Description                                       | Pass If                                                           | Fail If                                |
|-----------|---------------------------------------------------|-------------------------------------------------------------------|----------------------------------------|
| <b>C1</b> | Floquet tick permutes each orbit class internally | No cross-class transitions                                        | Any cross-class transition             |
| <b>C2</b> | $U_\theta$ preserves nearest deep hole identity   | $\text{nearest}(U_\theta(v_s(\varphi))) = s$ for all $s, \varphi$ | Any $s, \varphi$ where nearest changes |
| <b>C3</b> | Uniform phase rotation on 24-hole base            | $\delta\varphi = \theta \pm 10^{-6}$ for all $s$                  | Non-uniform shift                      |
| <b>C4</b> | Logical qubit encoding exists                     | Two distinguishable sections preserved by both operations         | No valid section pair found            |
| <b>C5</b> | Orbit-class sub-bundles are well-defined          | Each class has cyclic permutation, $U_\theta$ preserves class     | Non-cyclic or class-mixing             |

**Pass Condition:** C1, C2, and C3 pass. (C4 and C5 are exploratory.)

**Fail Condition:** C1, C2, or C3 fails.

## 6. Methodology

### Step 1: Verify Orbit-Class Internal Permutation (C1)

- For each deep hole  $s$  in each orbit class:
  - Apply one Floquet tick
  - Find nearest deep hole  $s'$
  - Record class transition:  $\text{class}(s) \rightarrow \text{class}(s')$
- Build transition matrix for classes
- Check if diagonal (no cross-class transitions)

### Step 2: Verify $U_\theta$ Preserves Nearest Deep Hole (C2)

- For each deep hole  $s$ :
  - Construct  $v = h_s$  (exact deep hole)
  - Apply  $U_\theta$  for  $\theta \in \{0.1, \pi/23, 2\pi/23, \pi/7, \pi/3, 1.0, 2.5\}$
  - Verify nearest deep hole is still  $s$
- For each deep hole  $s$  and phase  $\varphi \in [0, 2\pi)$  (grid of 100 values):
  - Construct  $v_s(\varphi)$  by rotating  $h_s$  in  $(0,23)$  plane to angle  $\varphi$
  - Apply  $U_\theta$
  - Verify nearest deep hole is still  $s$

### Step 3: Verify Uniform Phase Rotation (C3)

- For each  $s$  and  $\varphi$ :
  - Measure  $\varphi(v_s(\varphi))$
  - Apply  $U_\theta$
  - Measure  $\varphi(U_\theta(v_s(\varphi)))$

- Verify difference equals  $\theta$

**Step 4: Test Logical Qubit Encoding (C4)**

- Define two sections:
  - $|0_L\rangle$ : all phases  $\varphi_s = 0$  for all  $s$
  - $|1_L\rangle$ : alternating phases  $\varphi_s = 0$  for even  $s$ ,  $\varphi_s = \pi$  for odd  $s$
- Test  $U_\theta$  action: both sections should rotate uniformly
- Test Floquet action: sections should map to valid sections (possibly permuted base)
- Measure “orthogonality” via distance in phase pattern space

**Step 5: Analyze Orbit-Class Sub-Bundles (C5)**

- For each orbit class, compute the full permutation induced by one Floquet tick
- Decompose into cycles
- Verify  $U_\theta$  preserves class membership

**7. Expected Outcomes**

| Scenario                        | Verdict               | Meaning                                                                                                                               |
|---------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| C1–C3 pass, C4 passes           | <b>PASS (Strong)</b>  | The 24-hole base supports a genuine fiber bundle logical encoding with qubit structure.                                               |
| C1–C3 pass, C4 fails, C5 passes | <b>PASS (Partial)</b> | The 24-hole base is valid, but qubit encoding requires more sophisticated section design (e.g., using orbit-class sub-bundles).       |
| C1 or C2 fails                  | <b>FAIL</b>           | The 24-hole base is also incorrect. Need to search for a completely different base space (e.g., E8 root structure, 24-cell vertices). |
| C1 and C2 pass, C3 fails        | <b>FAIL</b>           | The base is preserved but $U_\theta$ is not a uniform fiber rotation. The fiber bundle interpretation is wrong.                       |

**8. Connection to Framework History**

| RC            | Contribution                                                   | Relevance to RC-139                                        |
|---------------|----------------------------------------------------------------|------------------------------------------------------------|
| <b>RC-138</b> | Failed 6-axis base, confirmed fiber coordinate exists          | Starting point —we know the fiber works, need correct base |
| <b>RC-137</b> | $U_\theta$ preserves nearest deep hole, uniform phase rotation | H3 verified —fiber is well-defined                         |

Table 4 – continued

| RC             | Contribution                                    | Relevance to RC-139                                           |
|----------------|-------------------------------------------------|---------------------------------------------------------------|
| <b>RC-134</b>  | DH23 fixed point, Class D genuine basins        | Class E is trivial; Class D has 2 holes —smallest sub-bundle  |
| <b>RC-130b</b> | 5-fold axis locality                            | Geometric structure exists but is not the dynamical base      |
| <b>RC-126</b>  | Orbit class partition (A:8, B:7, C:6, D:2, E:1) | The dynamical partition — candidate for sub-bundle structure  |
| <b>RC-124</b>  | Color sequence injectivity                      | Each deep hole has unique fingerprint —supports 24-point base |
| <b>RC-122</b>  | Icosahedron embedding                           | 12 vertices, 6 axes —geometric but not dynamical              |
| <b>RC-116</b>  | Dimensional cascade (24D $\rightarrow$ 3D)      | The 3D projection is a “shadow”; true structure is in 24D     |
| <b>RC-110</b>  | Floquet tick infrastructure                     | The engine that defines the dynamics —preserves orbit classes |

## 9. Honest Limitations

1. **The 24-hole base is large.** A section assigns a phase to each of 24 points, giving a 24-dimensional continuous space. A qubit encoding requires finding a 2-dimensional subspace — this may be difficult or arbitrary.
2. **The Floquet tick’s permutation structure is complex.** It may not be a simple cyclic permutation within each class. The permutation may depend on the tick number  $t$  (since  $H_L$  is applied every 11th tick).
3. **The “nearest deep hole” definition is metric-dependent.** We use Euclidean distance in 24D. Other metrics (e.g., Hamming distance on signs, angle in 3D projection) might give different results.
4. **The logical qubit test (C4) is exploratory.** We do not have a pre-registered specific section design. The alternating-phase design is a natural first guess but may not be optimal.
5. **The threshold  $10^{-6}$  remains arbitrary.** Sensitivity analysis should be performed.
6. **The framework uses real vectors, not complex amplitudes.** The “phase” is a real angle in the (0,23) plane, not a quantum phase. The logical “qubit” may not have standard quantum properties (superposition, entanglement).

## 10. Next Steps

| Outcome              | Next Cycle                                                                                                       |
|----------------------|------------------------------------------------------------------------------------------------------------------|
| <b>PASS (Strong)</b> | RC-140: Construct explicit magic state circuit for $U_\theta$ on the 24-hole fiber bundle; test logical gate set |

Table 5 – continued

| Outcome               | Next Cycle                                                                                                       |
|-----------------------|------------------------------------------------------------------------------------------------------------------|
| <b>PASS (Partial)</b> | RC-139b: Refine section design using orbit-class sub-bundles; test if Class B (7 holes) supports a qubit         |
| <b>FAIL</b>           | RC-139c: Search for a different base space entirely –E8 root system, 24-cell vertices, or cocode coset structure |

## 11. Pre-Registered Commitment

The author commits to:

1. **Report all results honestly**, including failures and null results.
2. **Not move goalposts**. If C1–C3 fail, the verdict is FAIL, and RC-139c will be executed.
3. **Not introduce post-hoc hypotheses**. The 5 candidate base spaces were identified before execution.
4. **Use only the pre-registered methodology**. No additional tests will be invented to rescue a failing hypothesis.
5. **Acknowledge limitations**. The 24-hole base may be too large, the metric arbitrary, and the qubit encoding non-quantum.

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**END OF PRE-REGISTRATION**